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Inference for unit roots in dynamic panels where the time dimension is fixed

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Abstract

This paper derives similar, asymptotic unit root tests for first-order autoregressive panel data models, assuming that the time dimension of the panel is fixed. It is shown that the limiting distributions of the test statistics are normal. The assumption that the time dimension is fixed allows us to derive analytical expressions for the moments of the distributions. Similarity with respect to the initial conditions of the data generating process is achieved by including fixed effect dummy variables in the regression model, while similarity with respect to fixed effects in the data generating process is achieved by including a linear deterministic trend for each individual unit of the panel. When fixed effects or individual trends are included as regressors the *least squares* estimator of the autoregressive parameter is inconsistent and thus the test statistics must be appropriately adjusted. Monte Carlo evidence suggests that the proposed tests have empirical size that is very close to the nominal five percent level and substantially more power than the corresponding unit root tests for the single time series case. © 1999 Elsevier Science S.A. All rights reserved.

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1. Introduction

Substantial research effort has now been devoted to developing inference techniques for detecting the presence of a unit autoregressive root in time series

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data. The extension of these techniques to the analysis of panel data is of considerable interest in order to study the dynamic behaviour of a number of economic units (such as individuals, firms or countries) over time (Hsiao, 1986; Baltagi, 1995). Unit root testing procedures for dynamic panel data models have been recently suggested by Quah (1992, 1994), Levin and Lin (1993), and Im et al. (1995).

The tests of Quah (1992, 1994) are limited to the case of panel data without fixed effects, and assume that both the cross-section dimension (N) and the time dimension (T) of the panel go to infinity ($\to \infty$) at the same rate, that is the ratio N/T is constant. Levin and Lin (1993) consider the more general model with fixed effects and individual deterministic trends, and allow for heterogeneous serially correlated errors. They assume that both N and T go to infinity, but that T increases faster than N, with $N/T \to 0$. They show that the inclusion of fixed effects or individual trends in the regression model creates a downward bias in the least squares estimator of the autoregressive parameter, and thus the tests statistics must be appropriately adjusted. Finally, Im et al. (1995) consider the same class of models and propose a test based on the average of N individual Augmented Dickey–Fuller statistics. They assume that both N and T go to infinity, but with the weaker assumption that $\sqrt{N}/T \to 0$. For the case that the error term is serially uncorrelated and independent across N, Im et al. (1995) demonstrate that their tests are also valid for the case that T is fixed.[1] All these test statistics have normal limiting distributions, whose analytical moments can be derived under the assumption that N and T go to infinity. However, in order for the more general tests of Levin and Lin (1993) and Im et al. (1995) to be made operational in small T samples, the authors suggest the use of simulated, rather than analytical values of the distribution moments.

In this paper, we develop asymptotic unit root tests for first-order autoregressive panel data models with serially uncorrelated errors, under the assumption that $N \to \infty$, while T is fixed. The assumption that T is fixed seems natural considering that in most empirical studies the number of cross-section units is typically large relative to the number of time series observations for each cross-section unit (see, for instance, Chamberlain, 1984). In contrast with the unit root tests discussed above, this assumption enables us to derive analytical expressions of the moments of the distributions of the tests by applying standard (classical) asymptotic theory across N (Anderson, 1978).

The tests derived in this paper are based on the normalised least squares (LS) estimators of the autoregressive coefficient and allow for fixed effects and

[1] The t-ratio test of Im et al. (1995) is valid for T greater than five when only an intercept is included in each Dickey–Fuller regression and for T greater than six when a trend is added. Their likelihood ratio test is valid for T greater than nine and T greater than ten, respectively.

individual deterministic trends. The tests are similar in the sense that their distribution under the null hypothesis is invariant with respect to the nuisance parameters of the data generating process (DGP) even in finite samples (Kiviet and Phillips, 1990, 1992). Similarity with respect to the initial conditions of the data generating process is achieved by the inclusion of fixed effects in the regression model, while similarity with respect to the fixed effects of the data generating process is achieved by the inclusion of the individual deterministic trends.

We show that the limiting distributions of the test statistics are normal and that their convergence rate is the same as for the case of stationary panel data, namely $\sqrt{N}$. The normality result does not require T to grow large. When fixed effects or individual deterministic trends are included in the regression the LS estimators are asymptotically biased downwards, as in the much studied case of stationary dynamic panels (see, inter alia, Nickell, 1981; Hsiao, 1986; Blundell and Smith, 1991; Sevestre and Trognon, 1992). In contrast with the stationary case, however, the bias remains even as T grows large (see also Levin and Lin, 1993). Where appropriate, the normalised coefficient statistic must be first corrected for the inconsistency of the LS estimator in order to yield a test statistic with a standard normal density.[2] We derive analytical expressions for both the inconsistency of the LS estimators when fixed effects or individual deterministic trends are included in the regression model, and the asymptotic variance of the resulting test statistics. Comparisons are made with the corresponding moments of the test statistics introduced by Quah (1994) and Levin and Lin (1993), who derive analogous results for the case that T grows large.

Monte Carlo experiments are used to evaluate the finite sample performance of the proposed tests, and to investigate the consequences for inference in finite samples of assuming that T is asymptotic rather than fixed. The results of these experiments indicate that the normal distribution accurately approximates the empirical distribution of the test statistics. Furthermore the proposed tests have substantially more power to discriminate between stationary and non-stationary processes than the corresponding unit root tests using single time series. It is shown that when fixed effects or individual trends are included in the regression model, the assumption that T is asymptotic yields a test which has substantially reduced power in small T samples.

The paper is organised as follows. Section 2 derives the test statistics based on the LS estimators of the autoregressive coefficient, and their limiting

[2] In the stationary case, generalised method of moments estimation is frequently employed to give consistent estimates of the autoregressive coefficient. However, this approach is not appropriate in the non-stationary case as the necessary moment conditions (over time) between the level of the series and the disturbance term are not satisfied (see, for instance, Ahn and Schmidt, 1995).

distributions. Section 3 investigates the finite sample size and power of the proposed panel unit root tests by means of Monte Carlo simulation. Finally, Section 4 offers some concluding remarks.

2. The test statistics and their limiting distributions

Following single time series unit root testing procedures (see Dickey and Fuller (1981), Phillips and Perron (1988), or for a survey, Stock (1994)), we consider the following three models.[3]

$$y_{it} = \varphi y_{it-1} + v_{it}, \qquad i = 1, \ldots, N; \quad t = 1, \ldots, T, \tag{1a}$$

$$y_{it} = \alpha_i + \varphi y_{it-1} + v_{it}, \tag{1b}$$

$$y_{it} = \alpha_i + \beta_i t + \varphi y_{it-1} + v_{it}. \tag{1c}$$

For each model, the null hypothesis is that there is a unit root present in the DGP, that is $\varphi = 1$. Under the alternative hypothesis, it is assumed that $|\varphi| < 1$. The appropriateness of each model depends, in part, on what is known about the DGP, and therefore how the null and alternative hypotheses are to be formulated. Model (1a) is the simple case of the homogeneous panel. Model (1b) is a unit root process with heterogeneous drift parameters under the null hypothesis, and a stationary process with heterogeneous intercepts under the alternative hypothesis. Model (1c), which includes heterogeneous fixed effects and individual trends, provides a test with better power to distinguish between the null hypothesis that each series follows a random walk with drift, and the alternative hypothesis that each series is stationary around a deterministic trend (Phillips and Perron, 1988). It will be seen that a second consideration in the choice between the Models (1a)–(1c) is that of similarity with respect to the nuisance parameters of the DGP.

Unit root test statistics have typically taken two forms in the single time series context. The first is the normalised coefficient statistic, which is based directly on the LS estimate of the autoregressive coefficient. The other is the t-ratio statistic. In this paper, we concentrate on the normalised coefficient statistic.

[3] These models correspond to those used by Levin and Lin (1993) and require that the autoregressive coefficient is homogeneous under both the null and alternative hypotheses, since they pool the data over the cross-section dimension. Im et al. (1995), who consider Models (1b) and (1c), do not pool the data, but instead base their test on N separate unit root regressions. This allows for heterogeneity in the autoregressive parameter under the alternative hypothesis. Quah (1994) considers only Model (1a) and assumes that the autoregressive parameter is homogeneous.

Under the null hypothesis that $\varphi = 1$, the LS estimators of φ in Models (1a)–(1c) satisfy

$$\hat{\varphi} - 1 = \left[\sum_{i=1}^{N} y'_{i,-1} Q_T y_{i,-1} \right]^{-1} \cdot \left[\sum_{i=1}^{N} y'_{i,-1} Q_T v_i \right] \tag{2}$$

where $y'_{i,-1} = (y_{i0}, \ldots, y_{iT-1})$, $v'_i = (v_{i1}, \ldots, v_{iT})$ and Q_T is the appropriate $(T \times T)$ indempotent transformation matrix. For Model (1a), $Q_T = I_T$, where I_T is the $(T \times T)$ identity matrix. For Model (1b), $Q_T = M_T = (I_T - (1/T)e_T e'_T)$ and e_T is the unit column vector of length T. For Model (1c), $Q_T = I_T - Z_T(Z'_T Z_T)^{-1} Z'_T$ with $Z_T = (e_T, \tau_T)$ and $\tau'_T = (1, 2, \ldots, T)$. The three LS estimators of φ corresponding to regression Models (1a)–(1c) are the least squares pooled estimator ($\hat{\varphi}_{\mathrm{LSP}}$), the least squares dummy variable estimator ($\hat{\varphi}_{\mathrm{LSDV}}$) and the least squares dummy variable with trend estimator ($\hat{\varphi}_{\mathrm{LSDVT}}$).

In order to derive the limiting distribution of the test statistics we must be precise about the nature of the series $\{v_{it}\}$, the initial conditions, y_{i0}, and the individual effects, α_i. In what follows, we maintain the following assumptions.

Assumptions. (a) $\{v_{it}; i = 1, \ldots, N, t = 1, \ldots, T\}$ is a series of independently and identically normally distributed (n.i.i.d.) random variables having $\mathrm{E}(v_{it}) = 0$ and $\mathrm{Var}(v_{it}) = \sigma_v^2 < \infty$, for all i and t; (b) the initial values, y_{i0}, are fixed; (c) the individual effects, α_i, are fixed.

Assumptions (a)–(c) imply that v_{it} and $y_{i,t-1}$, and their product $y_{i,t-1}v_{it}$, are independent across i.[4] This enables us to derive the limiting distributions of the normalised coefficient statistics for Models (1a)–(1c) by applying a Law of Large Numbers (LLN) and the Linderberg-Levy version of the Central Limit Theorem (CLT).[5] Assumption (a) implies that v_{it} is serially uncorrelated and thus makes the test comparable with the Dickey–Fuller test for single time series.

Assumption (b) requires that the initial conditions are observable constants, which implies that y_{i0} is uncorrelated with the sequence $\{v_{it}\}$. This is a reasonable assumption when the decision to start sampling is not arbitrary and

[4] The independence assumption can be relaxed to allow for time-varying aggregate effects in the data. These effects can be removed by including time-specific dummy variables in each regression model, or equivalently, by first subtracting the cross-section mean from the data. Removing aggregate effects in this way does not affect the LS estimators or the limiting distributions of the test statistics (O'Connell, 1997; Levin and Lin, 1993).

[5] Note that the normality assumption is not required to satisfy the conditions of the Linderberg–Levy CLT. However, this assumption greatly simplifies the testing procedure. As can be seen from the Appendix, the limiting distributions of the test statistics can be derived by relaxing the assumption of normality, but this requires a consistent estimator of the kurtosis coefficient of v_{it}.

therefore reflects the initial conditions of the panel at that particular point in time. It is also an appropriate assumption when the regression model includes fixed effects (see, for instance, Anderson and Hsiao, 1981, 1982). Evans and Savin (1984) also note that this assumption is suitable when the model is non-stable and thus does not converge in probability towards an unconditional mean. Similarly, Assumption (c) implies that α_i is uncorrelated with the sequence $\{v_{it}\}$.

We now present the limiting distributions of the normalised coefficient statistics for Models (1a)–(1c). In order to compare our results with those of Quah (1994) and Levin and Lin (1993), we assume that $y_{i0} = 0$ in Model (1a).

Theorem 1. If Assumption (*a*) *holds and* $y_{i0} = 0$ *for all i, then under the null hypothesis that* $\varphi = 1$,

$$\sqrt{N}(\hat{\varphi}_{\mathrm{LSP}} - 1) \xrightarrow{L} N(0, C_1) \tag{3}$$

where $C_1 = 2\{T(T-1)\}^{-1}$, and $\xrightarrow{L}$ denotes convergence in distribution.

The proof of Theorem 1 is given in the appendix, but it is instructive to sketch its outline here. The limiting distribution of $\sqrt{N}(\hat{\varphi}_{\mathrm{LSP}} - 1)$ is obtained by writing Eq. (2) as

$$(\hat{\varphi}_{\mathrm{LSP}} - 1) = \sum_{i=1}^{N} \sum_{t=1}^{T} y_{it-1} v_{it} \Big/ \sum_{i=1}^{N} \sum_{t=1}^{T} y_{it-1}^2 \tag{4}$$

and noting that for fixed T, $\sum_{t=1}^{T} y_{iT-1} v_{it}$ is a random variable, independently and identically distributed across i by Assumptions (a) and (b), with zero mean and finite variance given by $(T(T-1)/2)\sigma_v^4$. As $N \to \infty$, with T fixed, the numerator of Eq. (4) converges at the rate $\sqrt{N}$ to a normal random variable by the Linderberg–Levy CLT. The denominator converges in probability at the rate N to a non-stochastic scalar, and so the result of Theorem 1 follows by appropriately scaling the numerator and denominator of Eq. (4) and applying the Cramer–Wold lemma. It should be noted that the normality result does not require T to grow large.

Theorem 1 shows that the limiting distribution of the normalised coefficient statistic $\sqrt{N}(\hat{\varphi}_{\mathrm{LSP}} - 1)$ is normal with zero mean and variance given by C_1, which is decreasing in T. Therefore $\sqrt{N}(\hat{\varphi}_{\mathrm{LSP}} - 1)$, scaled by its asymptotic standard deviation, $C_1^{1/2}$, yields a test statistic which depends only on the estimated parameter and the known values of N and T, and thus can be readily used for statistical inference employing the critical values of the standard normal distribution.

The result of Theorem 1 is consistent with Breitung and Meyer (1994) who show that $\hat{\varphi}_{\mathrm{LSP}}$ is a $\sqrt{N}$ consistent estimator. Theorem 1 can also be compared with Quah (1994) and Levin and Lin (1993), both of whom analyse the limiting distribution of analogous statistics in the case that both N and T grow large. For sufficiently large T, Eq. (3) becomes $T\sqrt{N}(\hat{\varphi}_{\mathrm{LSP}}-1) \xrightarrow{L} \mathrm{N}(0, 2)$, which is the distribution given by Quah's Theorem 1 when the initial conditions are zero for all i, and by Theorem 4.7 of Levin and Lin for their test statistic Z_{δ}^{*} in the case that the error process is i.i.d. This asymptotic result implies that $\hat{\varphi}_{\mathrm{LSP}}$ converges at the rate $T\sqrt{N}$, which is higher than the convergence rate of the LS estimator in the stationary case. The higher convergence rate is due to the fact that as T grows large the time variability of y_{it} dominates its cross-section variability. The higher convergence rate with respect to T is known as 'super-consistency'.

It is interesting to note the consequences for inference in finite samples of assuming that T is asymptotic rather than fixed. It can be seen that the variance of the test statistic when T is assumed to be fixed is greater than its variance when T is assumed to be asymptotic by a factor of $T/(T-1)$. In small samples, therefore, the test based on the standardised test statistic assuming that T is asymptotic, using the nominal critical values of the standard normal distribution, will tend to reject the null hypothesis of a unit root too often, and hence the test will be oversized. The extent of this size distortion is investigated using Monte Carlo simulation in Section 3.

The results of Theorem 1 are specific to the case that y_{i0} is zero for all i, and thus may have limited practical application. Furthermore, if this assumption is relaxed the limiting distribution of Theorem 1 is no longer invariant to the nuisance parameters of the DGP, namely y_{i0} and σ_v^2.[6] However, a similar test statistic for the autoregressive parameter of Model (1a), which is invariant to the nuisance parameters of the DGP under the null hypothesis, even for finite samples, can be achieved by including N fixed effect dummy variables as redundant regressors in the regression model, that is by estimating, instead, Model (1b). The limiting distribution of the normalised bias statistic of $\hat{\varphi}_{\mathrm{LSDV}}$ is given by the following theorem.

Theorem 2. If Assumption (a) holds and $\alpha_i = 0$ for all i, then under the null hypothesis that $\varphi = 1$,

$$\sqrt{N}(\hat{\varphi}_{\mathrm{LSDV}} - 1 - B_2) \xrightarrow{L} \mathrm{N}(0, C_2) \tag{5}$$

where $B_2 = \mathrm{plim}_{N\to\infty}(\hat{\varphi}_{\mathrm{LSDV}} - 1) = -3(T+1)^{-1}$ and $C_2 = \{3(17T^2 - 20T + 17)\}\{5(T-1)(T+1)^3\}^{-1}$.

[6] Harris and Tzavalis (1996) give the limiting distribution of $\hat{\varphi}_{\mathrm{LSP}}$ for the more general case that $y_{i0} \neq 0$.

The proof of Theorem 2 is given in Appendix.

Theorem 2 demonstrates that for $N \to \infty$ with T fixed, $\hat{\varphi}_{\mathrm{LSDV}}$ is an inconsistent estimator, as in the stationary case that $|\varphi| < 1$. The inconsistency of $\hat{\varphi}_{\mathrm{LSDV}}$, given by B_2, is negative and depends only on T. The inconsistency arises from the elimination of an unknown constant from each observation of the panel, which yields a correlation between the mean of $y_{i,\,t-1}$ and v_{it} for each i (Hsiao, 1986). Correcting for the inconsistency of $\hat{\varphi}_{\mathrm{LSDV}}$ yields a statistic which has a normal limiting distribution with asymptotic variance, C_2.

For sufficiently large T, Eq. (5) implies that $T\sqrt{N}(\hat{\varphi}_{\mathrm{LSDV}} - 1) + 3\sqrt{N} \xrightarrow{L} \mathrm{N}(0, \frac{51}{5})$, which is consistent with the distribution of the test statistic Z_δ^* given by Theorem 4.1 of Levin and Lin (1993) for $\alpha_i = 0$.[7] The term $3\sqrt{N}$ is a necessary adjustment for the inconsistency of $\hat{\varphi}_{\mathrm{LSDV}}$ as N and T go to infinity.[8] The inconsistency remains since it is not influenced by the cross-section dimension of the panel and the convergence rate (with respect to T) of C_2 is bigger than that of B_2. Thus, as noted by Levin and Lin (1993), in contrast to the case of stationary panel data, the presence of a unit root causes the fixed effects to influence the asymptotic distribution of the panel autoregressive estimator, even as both N and T become large.

As with Theorem 1, we can analyse the consequences for inference in finite samples of assuming that T is asymptotic rather than fixed. When T is assumed to be asymptotic, the adjustment term required for the inconsistency of φ_{LSDV} is larger than when T is assumed to be fixed (by a factor of $(T+1)/T$). The variance of the statistic $\sqrt{N}(\hat{\varphi}_{\mathrm{LSDV}} - 1 - B_2)$, when T is assumed to be asymptotic, is also larger (by a factor of $17(T-1)(T+1)^3/T^2(17T^2 - 20T + 17)$), and so the variance of the *standardised* test statistic, $C^{-1/2}\sqrt{N}(\hat{\varphi}_{\mathrm{LSDV}} - 1 - B_2)$, is smaller. Together, these have the effect of shifting the distribution of the standardised test statistic to the left, relative to when T is assumed to be fixed, by

$$3\sqrt{N}\left[\left(\frac{5(T-1)(T+1)^3}{3(T+1)^2(17T^2 - 20T + 7)}\right)^{1/2} - \left(\frac{5}{51}\right)^{1/2}\right],$$

[7] To obtain this result, and the corresponding result for Model (1c), the limiting distribution of the test statistic Z_δ^* of Levin and Lin must be adjusted for the inconsistency of the LS estimator of the autoregressive coefficient, rather than by the asymptotic mean of the numerator of the estimator. Then the variance of the limiting distribution of Z_δ^* becomes

$$\frac{\sigma_{m1}^2}{\mu_{m2}^2} + \left(\frac{\mu_{m1}}{\mu_{m2}}\right)^2 \frac{\sigma_{m2}^2}{\mu_{m2}^2},$$

where the moments for each model are computed by Levin and Lin (see their Table 1).

[8] This result is consistent with Bernard and Jones (1996) who consider panel data unit root tests when the a_i are random effects. Note that in this case the bias disappears as $N \to \infty$ and $T \to \infty$.

which increases the size of the test over the nominal level. However, since the variance of the standardised statistic is smaller when T is assumed to be asymptotic, the tails of the distribution are drawn in, which decreases the empirical size of the test. If the first of these effects dominates the second, then the test will be oversized and will reject the null hypothesis of a unit root too often when using the nominal critical values of the standard normal distribution. If the second effect is dominant, then the test will be undersized and hence have low power to reject the null hypothesis. The relative importance of these effects are investigated using Monte Carlo simulation in Section 3.

The results of Theorem 2 are specific to the case that $\alpha_i = 0$ for all i. In the more general case that $\alpha_i \neq 0$, the test statistic given by Theorem 2 will depend on the nuisance parameters of the DGP, namely α_i and σ_v^2.[9] However, similarity with respect to these nuisance parameters can be achieved by including individual deterministic trends in the regression model, that is by estimating instead regression Model (1c). In this case, the limiting distribution of the test statistic is given by the following theorem.

Theorem 3. If Assumption (a) holds and $\beta_i = 0$ for all i, then under the null hypothesis that $\varphi = 1$

$$\sqrt{N}(\hat{\varphi}_{\mathrm{LSDVT}} - 1 - B_3) \xrightarrow{L} \mathrm{N}(0, C_3) \tag{6}$$

where $B_3 = \mathrm{plim}_{N\to\infty}(\hat{\varphi}_{\mathrm{LSDVT}} - 1) = -15\{2(T+2)\}^{-1}$ *and* $C_3 = \{15(193T^2 - 728\mathrm{T} + 1147)\}\{112(T+2)^3(T-2)\}^{-1}$.

The proof of Theorem 3 is given in Appendix.

The inconsistency of $\hat{\varphi}_{\mathrm{LSDVT}}$, given by B_3, is larger than in the case of Theorem 2. This is due to the fact that now, in addition to an unknown constant, a linear trend is subtracted from each observation which leads to greater correlation between the mean of $y_{i,\,t-1}$ and v_{it} for each i.

For T sufficiently large, Eq. (8) implies that $T\sqrt{N}(\hat{\varphi}_{\mathrm{LSDV}} - 1) + 7.5\sqrt{N} \xrightarrow{L} \mathrm{N}(0, 2895/112)$, which is consistent with the distribution of the test statistic Z_δ^* given by Theorem 4.1 of Levin and Lin (1993) for $\alpha_i \neq 0$ (see footnote 7). Note that in this case, since $B_3 > B_2$, the adjustment factor that is required for the inconsistency which remains as N and T go to infinity is bigger than that corresponding to Theorem 2.

[9] Harris and Tzavalis (1996) consider the case that $\alpha_i \neq 0$. When $N = 1$ and $T \to \infty$, this corresponds to Theorem 3.1 of West (1988).

We can again consider the consequences for inference in finite samples of assuming that T is asymptotic rather than fixed. As with Theorem 2, using the moments based on asymptotic T rather than fixed T to compute the test statistic has two counteracting effects on the size of the test. When T is assumed to be asymptotic, the magnitude of the adjustment for the inconsistency of φ_{LSDVT} is larger (by a factor of $(T+2)/T$). The variance of the statistic $\sqrt{N}(\hat{\varphi}_{\mathrm{LSDVT}}-1-B_2)$ is larger (by a factor of $15T^2(193T^2-728T+1147)/2895(T-2)(T+2)^3$) and so the variance of the *standardised* test statistic, $C^{-1/2}\sqrt{N}(\hat{\varphi}_{\mathrm{LSDVT}}-1-B_2)$ is smaller. Together, these have the effect of shifting the distribution of the standardised statistic to the left by

$$7.5\sqrt{N}\left[\left(\frac{112(T-2)(T+2)^3}{15(T+2)^2(193T^2-728T+1147)}\right)^{1/2}-\left(\frac{112}{2895}\right)^{1/2}\right],$$

which increases the size of the test over the nominal level. On the other hand, since the variance of the standardised statistic is smaller when T is assumed to be asymptotic, the tails of the distribution are drawn in, which decreases the empirical size of the test.

3. Simulation experiments

In this section we present two sets of Monte Carlo experiments. The first assesses the finite sample accuracy of the limiting distributions and the power of the test statistics derived in this paper. We perform these experiments for the normalised coefficient statistic corresponding to the simple case of Theorem 1 and the similar tests suggested by Theorems 2 and 3. The second set of experiments investigates the consequences for inference in finite samples of using the test statistics based on asymptotic T, rather than fixed T. This enables us to judge the appropriate size of T that can be reasonably considered to be asymptotic.

The empirical quantiles of the tests and their size for different combinations of N and T, using 5000 replications, are reported in Table 1a–c. The last column of the table gives the true size of a one-sided test with critical values taken from the standard normal distribution at the nominal 5% level of significance. For comparison, we also include in the tables the corresponding quantiles of the standard normal distribution. The null model is given in each table. In each experiment, the data are generated according to the null model with disturbance terms drawn independently from the standard normal distribution.

Table 1a–c show that the standard normal distribution accurately approximates the empirical distribution of the test statistics for the cases where N is large

relative to T, which is typical in most dynamic panels, or for sufficiently large N. Consequently, for these cases the tests have empirical size very close to the nominal 5% level. On the other hand, for panels of very small N and relatively large T the empirical distribution approaches the Dickey–Fuller density (Dickey and Fuller, 1981). This is because the time dimension of the panel in this case dominates the cross-section dimension. The slight size deterioration observed in Table 1a for small T and N (such as $\{T, N\} = \{5, 5\}$) may be due to the fact that for $\hat{\varphi}_{\mathrm{LSP}}$, finite ($T$) sample results depend on y_{i0} (Abadir, 1993).

The empirical power of the tests is reported in Table 2a–c. We generate data according to the three alternative hypotheses that $\varphi = 0.8$, $\varphi = 0.9$ and $\varphi = 0.95$ respectively, and tabulate both the size-adjusted power using the actual 5% critical values taken from Table 1a–c and the nominal power using the 5% critical values of the standard normal distribution. In all experiments y_{i0} was set to zero (as in Im et al., 1995). Using the size adjusted power enables us to better judge the relative performance of the three tests for small N, where the actual

Table 1a
Empirical quantiles of $C_1^{-1/2}\sqrt{N}(\hat{\phi}_{\mathrm{LSP}} - 1)$ for $(y_{i0}, \varphi) = (0, 1)$

T	N	1%	5%	10%	50%	90%	95%	99%	Size
5	5	− 3.76	− 2.52	− 1.92	− 0.14	1.03	1.34	1.91	0.13
5	10	− 3.31	− 2.23	− 1.71	− 0.12	1.10	1.39	2.03	0.10
5	25	− 2.88	− 1.95	− 1.49	− 0.05	1.14	1.46	2.06	0.08
5	50	− 2.89	− 1.84	− 1.39	− 0.04	1.17	1.48	2.07	0.07
5	100	− 2.60	− 1.82	− 1.38	− 0.01	1.18	1.51	2.16	0.07
10	5	− 4.37	− 2.71	− 2.04	− 0.21	0.95	1.20	1.71	0.14
10	10	− 3.75	− 2.27	− 1.71	− 0.11	1.02	1.31	1.80	0.11
10	25	− 3.06	− 2.00	− 1.54	− 0.07	1.10	1.40	1.95	0.08
10	50	− 2.86	− 1.91	− 1.45	− 0.04	1.14	1.47	2.04	0.07
10	100	− 2.60	− 1.83	− 1.43	− 0.04	1.18	1.49	2.08	0.07
25	5	− 4.45	− 2.85	− 2.06	− 0.24	0.92	1.18	1.60	0.15
25	10	− 3.77	− 2.43	− 1.82	− 0.15	1.00	1.31	1.81	0.12
25	25	− 3.10	− 2.05	− 1.55	− 0.09	1.07	1.38	1.91	0.09
25	50	− 2.82	− 1.95	− 1.51	− 0.06	1.15	1.44	2.00	0.08
25	100	− 2.60	− 1.84	− 1.43	− 0.05	1.17	1.47	2.02	0.07
50	5	− 4.94	− 2.94	− 2.14	− 0.22	0.90	1.17	1.62	0.15
50	10	− 3.99	− 2.41	− 1.79	− 0.13	1.02	1.26	1.72	0.12
50	25	− 3.13	− 2.18	− 1.60	− 0.09	1.09	1.38	1.86	0.09
50	50	− 2.67	− 1.85	− 1.44	− 0.09	1.10	1.40	1.95	0.07
50	100	− 2.72	− 1.78	− 1.35	− 0.05	1.15	1.49	2.11	0.06
$N(0, 1)$		− 2.33	− 1.65	− 1.28	0.00	1.28	1.65	2.33	

Table 1b
Empirical quantiles of $C_2^{-1/2}\sqrt{N}(\hat{\phi}_{\mathrm{LSDV}} - 1 - B_2)$ for $(\alpha_i, \varphi) = (0, 1)$

T	N	1%	5%	10%	50%	90%	95%	99%	Size
5	5	− 2.82	− 2.03	− 1.62	− 0.14	1.14	1.50	2.14	0.09
5	10	− 2.83	− 1.97	− 1.54	− 0.09	1.14	1.51	2.06	0.08
5	25	− 2.60	− 1.85	− 1.44	− 0.06	1.21	1.54	2.12	0.07
5	50	− 2.55	− 1.79	− 1.40	− 0.04	1.26	1.60	2.20	0.07
5	100	− 2.45	− 1.77	− 1.40	− 0.04	1.22	1.57	2.27	0.06
10	5	− 3.27	− 2.24	− 1.68	− 0.19	1.03	1.33	1.88	0.10
10	10	− 2.97	− 2.02	− 1.57	− 0.12	1.09	1.40	1.96	0.09
10	25	− 2.81	− 1.98	− 1.51	− 0.09	1.18	1.52	2.13	0.08
10	50	− 2.66	− 1.88	− 1.48	− 0.05	1.22	1.60	2.18	0.07
10	100	− 2.60	− 1.82	− 1.44	− 0.04	1.24	1.60	2.18	0.06
25	5	− 3.44	− 2.36	− 1.81	− 0.21	0.95	1.24	1.67	0.12
25	10	− 3.14	− 2.14	− 1.66	− 0.12	1.08	1.35	1.86	0.10
25	25	− 3.00	− 1.97	− 1.55	− 0.06	1.19	1.51	2.99	0.08
25	50	− 2.75	− 1.96	− 1.52	− 0.06	1.24	1.57	2.11	0.08
25	100	− 2.68	− 1.81	− 1.44	− 0.05	1.23	1.60	2.33	0.07
50	5	− 3.58	− 2.41	− 1.83	− 0.17	0.98	1.25	1.67	0.12
50	10	− 3.15	− 2.09	− 1.65	− 0.15	1.00	1.32	1.85	0.10
50	25	− 3.02	− 1.99	− 1.53	− 0.10	1.14	1.44	2.17	0.08
50	50	− 2.64	− 1.86	− 1.44	− 0.07	1.19	1.53	2.12	0.07
50	100	− 2.67	− 1.84	− 1.43	− 0.06	1.25	1.61	2.20	0.07
$N(0, 1)$		− 2.33	− 1.65	− 1.28	0.00	1.28	1.65	2.33	

bias of the LS estimators may not be well approximated by the inconsistency given by Theorems 1–3. As N grows large, the size adjusted power tends to the nominal power. For the test given by Theorem 2 the data are generated assuming that the intercept, α_i, is set to zero for all i under both the null and alternative hypotheses, ensuring that the mean of Δy_{it} is the same in both cases. For the test given by Theorem 3 the data are generated according to $y_{it} = \alpha_i + \beta_i t + \varphi y_{it-1} + u_{it}$, where α_i is drawn once for each i from the standard normal distribution and $\beta_i = \alpha_i(1 - \varphi)$. This specification ensures that the intercept parameter is the same under both the null and the alternative hypotheses and thus enables us to evaluate the ability of the test to reject the null hypothesis of a unit root against the trend stationary alternative, where the rate of trend is the same under both the null and alternative hypotheses.

As expected, the estimated power of the tests is very low for values of φ very close to unity and increases monotonically with both N and T. For all three

Table 1c
Empirical quantiles of $C_3^{-1/2}\sqrt{N}(\hat{\varphi}_{\mathrm{LSDVT}} - 1 - B_3)$ for $(\beta_i, \varphi) = (0, 1)$ and α_i drawn from $N(0, 1)$ once for each i

T	N	1%	5%	10%	50%	90%	95%	99%	Size
5	5	− 2.37	− 1.71	− 1.36	− 0.12	1.21	1.58	2.48	0.06
5	10	− 2.28	− 1.65	− 1.30	− 0.09	1.24	1.62	2.33	0.05
5	25	− 2.31	− 1.67	− 1.36	− 0.05	1.24	1.55	2.31	0.05
5	50	− 2.43	− 1.66	− 1.30	− 0.03	1.25	1.61	2.23	0.05
5	100	− 2.29	− 1.63	− 1.26	− 0.02	1.23	1.60	2.30	0.05
10	5	− 2.61	− 1.84	− 1.45	− 0.14	1.10	1.45	2.01	0.07
10	10	− 2.48	− 1.77	− 1.41	− 0.10	1.15	1.46	2.05	0.06
10	25	− 2.45	− 1.78	− 1.39	− 0.06	1.19	1.52	2.18	0.06
10	50	− 2.49	− 1.76	− 1.36	− 0.05	1.21	1.57	2.17	0.06
10	100	− 2.47	− 1.71	− 1.33	− 0.05	1.22	1.59	2.21	0.05
25	5	− 2.92	− 2.02	− 1.58	− 0.14	1.00	1.30	1.81	0.09
25	10	− 2.80	− 1.95	− 1.52	− 0.08	1.08	1.40	1.87	0.08
25	25	− 2.63	− 1.77	− 1.40	− 0.06	1.14	1.46	2.00	0.06
25	50	− 2.57	− 1.76	− 1.38	− 0.06	1.17	1.51	2.09	0.06
25	100	− 2.44	− 1.75	− 1.40	− 0.03	1.26	1.57	2.21	0.06
50	5	− 3.03	− 2.06	− 1.57	− 0.16	0.97	1.30	1.84	0.09
50	10	− 2.82	− 1.97	− 1.54	− 0.09	1.09	1.37	1.96	0.09
50	25	− 2.61	− 1.81	− 1.42	− 0.06	1.13	1.44	2.01	0.07
50	50	− 2.46	− 1.72	− 1.35	− 0.05	1.19	1.52	2.15	0.06
50	100	− 2.48	− 1.73	− 1.33	0.04	1.22	1.57	2.18	0.06
$N(0, 1)$		− 2.33	− 1.65	− 1.28	0.00	1.28	1.65	2.33	

tests, as N grows large the power approaches its maximum value of unity even for very small T. The power improves faster as T increases (with N given) than as N increases (with T given). It is also interesting to note that the power achieves its maximum value for sizes of T which are much smaller than for the corresponding single time series Dickey–Fuller tests (see, for instance, Dickey and Fuller, 1981), suggesting that for given T, the power is an increasing function of N due to the increased information content of the sample arising from its cross-section variation.

The deterioration of the estimated power of the normalised coefficient statistics associated with Theorems 2 and 3, compared with Theorem 1, is a consequence of including redundant regressors in the unit root regressions to achieve test similarity with regard to the nuisance parameters of the DGP (Kiviet and Phillips, 1992). The loss in power is the price paid for the property of similarity.

Table 2a
Empirical power of $C_1^{-1/2}\sqrt{N}(\hat{\varphi}_{\mathrm{LSP}} - 1)$

		$\varphi = 0.8$		$\varphi = 0.9$		$\varphi = 0.95$	
T	N	Size adjusted	Nominal	Size adjusted	Nominal	Size adjusted	Nominal
5	5	0.24	0.47	0.12	0.26	0.08	0.19
5	10	0.47	0.64	0.19	0.33	0.10	0.19
5	25	0.86	0.92	0.39	0.49	0.16	0.23
5	50	0.99	0.99	0.65	0.72	0.25	0.31
5	100	1.00	1.00	0.90	0.93	0.40	0.47
10	5	0.63	0.88	0.25	0.52	0.12	0.31
10	10	0.94	0.98	0.49	0.68	0.20	0.34
10	25	1.00	1.00	0.89	0.94	0.40	0.53
10	50	1.00	1.00	0.99	0.99	0.67	0.76
10	100	1.00	1.00	1.00	1.00	0.93	0.95
25	5	0.99	1.00	0.79	0.96	0.33	0.66
25	10	1.00	1.00	0.99	0.99	0.64	0.84
25	25	1.00	1.00	1.00	1.00	0.97	0.99
25	50	1.00	1.00	1.00	1.00	1.00	1.00
25	100	1.00	1.00	1.00	1.00	1.00	1.00
50	5	1.00	1.00	0.99	1.00	0.77	0.96
50	10	1.00	1.00	1.00	1.00	0.99	0.99
50	25	1.00	1.00	1.00	1.00	1.00	1.00
50	50	1.00	1.00	1.00	1.00	1.00	1.00
50	100	1.00	1.00	1.00	1.00	1.00	1.00

The results of the second set of experiments are reported in Table 3. The table gives the size and power of the tests suggested by Theorems 1–3 using the 5% critical values of the standard normal distribution. The test statistics are constructed under the assumption that T is large. The power is considered against the alternative hypothesis that $\varphi = 0.95$.[10] The data are generated in the same way as for the first set of experiments.

For the test statistic implied by Theorem 1, Table 3 indicates that the assumption that T is asymptotic rather than fixed leads to a test that is oversized (cf. Table 1a). The size distortion is much more serious for small T, such as $T = \{5, 10\}$, but becomes negligible once T reaches about 25 observations, where both tests have approximately the same size. Consistent with the increase

[10] The estimates of the power for the other values of φ are not reported for reasons of space.

Table 2b
Empirical power of $C_2^{-1/2}\sqrt{N}(\hat{\varphi}_{\mathrm{LSDV}} - 1 - B_2)$

		$\varphi = 0.8$		$\varphi = 0.9$		$\varphi = 0.95$	
T	N	Size adjusted	Nominal	Size adjusted	Nominal	Size adjusted	Nominal
5	5	0.18	0.29	0.10	0.18	0.07	0.13
5	10	0.30	0.41	0.15	0.22	0.09	0.14
5	25	0.62	0.69	0.28	0.39	0.13	0.17
5	50	0.90	0.91	0.47	0.51	0.20	0.23
5	100	1.00	1.00	0.72	0.76	0.31	0.35
10	5	0.30	0.50	0.15	0.30	0.09	0.19
10	10	0.58	0.70	0.28	0.40	0.14	0.23
10	25	0.90	0.94	0.54	0.64	0.24	0.34
10	50	0.99	0.99	0.82	0.87	0.40	0.48
10	100	1.00	1.00	0.98	0.98	0.66	0.71
25	5	0.80	0.93	0.37	0.59	0.18	0.36
25	10	0.98	0.99	0.66	0.79	0.31	0.47
25	25	1.00	1.00	0.96	0.97	0.63	0.73
25	50	1.00	1.00	1.00	1.00	0.88	1.00
25	100	1.00	1.00	1.00	1.00	0.99	0.99
50	5	1.00	1.00	0.76	0.92	0.35	0.58
50	10	1.00	1.00	0.98	0.99	0.66	0.78
50	25	1.00	1.00	1.00	1.00	0.95	0.97
50	50	1.00	1.00	1.00	1.00	1.00	1.00
50	100	1.00	1.00	1.00	1.00	1.00	1.00

in size, for small T the power of the test is greater than when T is assumed to be fixed (cf. Table 2a).

For the test statistics associated with Theorems 2 and 3, Table 3 shows that the assumption that T is asymptotic rather than fixed yields a test which is substantially undersized (cf. Table 1b and c). It was noted in Section 2 that this assumption has two counteracting effects on the size of the test : the distribution of the standardised test statistic is shifted to the left (increasing the size of the test), but the variance of the standardised statistic is smaller (reducing the size of the test). The results of Table 3 indicate that the variance effect dominates the mean shift effect, resulting in a net reduction in the size of the test. The consequence of this is a considerable decrease in the power of the test to reject the null hypothesis (cf. Table 2b and c). The differences in the size and the power of the tests based on fixed T and asymptotic T disappear for values of T greater than about 50 observations.

Table 2c
Empirical power of $C_3^{-1/2}\sqrt{N}(\hat{\varphi}_{\mathrm{LSDVT}} - 1 - B_3)$

		$\varphi = 0.8$		$\varphi = 0.9$		$\varphi = 0.95$	
T	N	Size adjusted	Nominal	Size adjusted	Nominal	Size adjusted	Nominal
5	5	0.05	0.06	0.05	0.05	0.05	0.05
5	10	0.05	0.05	0.04	0.05	0.04	0.04
5	25	0.07	0.08	0.05	0.06	0.05	0.06
5	50	0.08	0.08	0.05	0.06	0.05	0.08
5	100	0.11	0.11	0.06	0.06	0.05	0.05
10	5	0.08	0.11	0.06	0.08	0.05	0.07
10	10	0.12	0.14	0.07	0.08	0.06	0.07
10	25	0.17	0.20	0.08	0.09	0.06	0.07
10	50	0.26	0.30	0.08	0.10	0.06	0.07
10	100	0.44	0.47	0.11	0.13	0.06	0.07
25	5	0.30	0.43	0.10	0.17	0.06	0.11
25	10	0.51	0.61	0.14	0.21	0.07	0.11
25	25	0.91	0.92	0.27	0.31	0.09	0.11
25	50	0.99	0.99	0.44	0.48	0.12	0.14
25	100	1.00	1.00	0.69	0.72	0.17	0.19
50	5	0.89	0.95	0.30	0.43	0.10	0.17
50	10	0.99	0.99	0.50	0.62	0.15	0.21
50	25	1.00	1.00	0.89	0.92	0.26	0.31
50	50	1.00	1.00	0.99	0.99	0.44	0.47
50	100	1.00	1.00	1.00	1.00	0.68	0.71

4. Conclusions

This paper introduces similar, asymptotic unit root tests for first order autoregressive panel data models where the time dimension of the panel is fixed, allowing for fixed effects and individual deterministic trends. The tests are based on the normalised LS estimator of the autoregressive coefficient, corrected where appropriate for its inconsistency. The inconsistency arises as a result of the inclusion of fixed effects and individual trends in the regression model, and is influenced only by T. The limiting distributions of the tests are shown to be normal. Comparisons are made with the limiting distributions of the corresponding test statistics given by Quah (1994) and Levin and Lin (1993), who consider the case that both N and T grow large. Monte Carlo experiments suggest that the standard normal density accurately approximates the empirical distributions of the tests, and that for sufficiently large N the tests have an empirical size very close to the nominal 5% level. The power performance of the tests against local alternatives is found to be considerably better than that of the

Table 3
Size and power of the tests at the nominal level 5% ($\varphi = 0.95$), assuming that T is large

		Statistic 1		Statistic 2		Statistic 3	
T	N	Size	Power	Size	Power	Size	Power
5	5	0.15	0.22	0.01	0.02	0.01	0.00
5	10	0.12	0.23	0.01	0.02	0.00	0.00
5	25	0.10	0.28	0.00	0.01	0.00	0.00
5	50	0.09	0.37	0.00	0.00	0.00	0.00
5	100	0.08	0.54	0.00	0.00	0.00	0.00
10	5	0.15	0.32	0.05	0.10	0.02	0.02
10	10	0.11	0.37	0.03	0.10	0.01	0.01
10	25	0.09	0.56	0.02	0.14	0.01	0.01
10	50	0.08	0.79	0.01	0.19	0.01	0.02
10	100	0.08	0.95	0.00	0.27	0.01	0.01
25	5	0.15	0.67	0.09	0.30	0.06	0.07
25	10	0.12	0.85	0.07	0.39	0.05	0.07
25	25	0.09	0.99	0.05	0.64	0.04	0.07
25	50	0.08	1.00	0.04	0.86	0.03	0.09
25	100	0.07	1.00	0.02	0.98	0.03	0.12
50	5	0.15	0.96	0.11	0.56	0.07	0.15
50	10	0.11	0.99	0.08	0.75	0.06	0.18
50	25	0.09	1.00	0.06	0.96	0.05	0.27
50	50	0.07	1.00	0.05	1.00	0.04	0.42
50	100	0.06	1.00	0.05	1.00	0.04	0.68

Statistic 1 is $2^{-1/2}T\sqrt{N}(\hat{\phi}_{\mathrm{LSP}} - 1)$, corresponding to Theorem 1.
Statistic 2 is $(5/51)^{-1/2}(T\sqrt{N}(\hat{\phi}_{\mathrm{LSDV}} - 1) + 3\sqrt{N})$, corresponding to Theorem 2.
Statistic 3 is $(112/2895)^{-1/2}(T\sqrt{N}(\hat{\phi}_{\mathrm{LSDVT}} - 1) + 7.5\sqrt{N})$, corresponding to Theorem 3.

single time series Dickey–Fuller tests and increases monotonically with both T and N. Furthermore, it is shown that the assumption that T is asymptotic yields a test which has substantially reduced power in small T samples.

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Appendix A.

This appendix contains the proofs of the theorems that are given in the main text. The derivations of the results are based on the Linderberg–Levy version of the Central Limit Theorem, the Strong Law of Large Numbers (SLLN), and the results of the following Lemma.

Lemma A.1. Let $\varepsilon_{it} = \sum_{t=1}^{t} v_{it}$, *then*

(a) $\sum_{t=1}^{T} \varepsilon_{it-1} = \sum_{t=1}^{T} (T-t) v_{it}$,

(b) $\sum_{t=1}^{T} \varepsilon_{it-1}^2 = \sum_{t=1}^{T} (T-t) v_{it}^2 + 2\sum_{t=1}^{T-1} \sum_{s=t+1}^{T} (T-s) v_{it} v_{is}$,

(c) $\sum_{t=1}^{T} \varepsilon_{it-1} v_{it} = \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} v_{it} v_{is}$,

(d) $\sum_{t=1}^{T} t\varepsilon_{it-1} = \sum_{t=1}^{T} ((T+t+1)(T-t)/2) v_{it}$,

(e) $\sum_{t=1}^{T} \varepsilon_{it-1} \sum_{t=1}^{T} v_{it} = \sum_{t=1}^{T} (T-t) v_{it}^2 + \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} (2T-s-t) v_{it} v_{is}$,

(f) $$\sum_{t=1}^{T} t\varepsilon_{it-1} \sum_{t=1}^{T} v_{it} = \sum_{t=1}^{T} ((T+t+1)(T-t)/2) v_{it}^2 + \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} ((T+t+1)(T-t) + (T+s+1)(T-s)/2) v_{it} v_{is},$$

(g) $$\sum_{t=1}^{T} t\varepsilon_{it-1} \sum_{t=1}^{T} t v_{it} = \sum_{t=1}^{T} ((T+t+1)(T-t)t/2) v_{it}^2 + \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} ((T+t+1)(T-t)s + (T+s+1)(T-s)t/2) v_{it} v_{is},$$

(h) $$\sum_{t=1}^{T} t\varepsilon_{it-1} \sum_{t=1}^{T} \varepsilon_{it-1} = \sum_{t=1}^{T} ((T+t+1)(T-t)^2/2) v_{it}^2 + \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} ((T-t)(T-s) \times (2T+t+s+2)/2) v_{it} v_{is}.$$

Proof. Results (a)–(d) follow immediately from the definition of ε_{it}, while results (e)–(h) follow by using (a)–(d) and the general result that, for any functions $f(\cdot)$ and $g(\cdot)$

$$\sum_{t=1}^{T} f(t)v_{it} \sum_{t=1}^{T} g(t)v_{it} = \sum_{t=1}^{T} f(t)g(t)v_{it}^2 + \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} [f(t)g(s) + f(s)g(t)]v_{it}v_{is}.$$

Proof of Theorem 1. Define $w_{iT} = \sum_{t=1}^{T} \varepsilon_{it-1}v_{it}$. Since by Assumption (a), v_{it} and ε_{it-1} are independent across i, and v_{it} are i.i.d. across i and t, $\{w_{iT}\}$ is an independent random sequence across i with mean zero and constant variance, given by $\sigma_{w_T}^2 = \sigma_v^4 \sum_{t=1}^{T}(t-1) = \sigma_v^4 T(T-1)/2 < \infty$. Applying the CLT, for $N \to \infty$, to the sum of $\{w_{iT}\}$ yields

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} \frac{w_{iT}}{\sigma_{w_T}} \xrightarrow{L} \mathrm{N}(0, 1) \tag{A.1}$$

The numerator of Eq. (4) can be expanded by recursive substitution to yield

$$\sum_{i=1}^{N} \sum_{t=1}^{T} y_{it-1}v_{it} = \sum_{i=1}^{N} \sum_{t=1}^{T} y_{i0}v_{it} + \sum_{i=1}^{N} w_{i,\,T} \tag{A.2}$$

which using Eq. (A.1) implies

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} \frac{1}{\sigma_{w_T} t=1} \sum^{T} y_{it-1}v_{it} \xrightarrow{L} \mathrm{N}(0, 1) \tag{A.3}$$

since $y_{i0} = 0$. Note that the last result implies that $\hat{\varphi}_{\mathrm{LSP}}$ is $\sqrt{N}$ consistent. Expanding the denominator in a similar fashion, and noting that v_{it} are independent across i, implies that

$$\operatorname*{plim}_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} y_{it-1}^2 = \sum_{t=1}^{T} (T-t) \operatorname*{plim}_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} v_{it}^2 = \frac{T(T-1)}{2} \sigma_v^2 \tag{A.4}$$

where

$$\operatorname*{plim}_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} v_{it}^2 = \sigma_v^2$$

by the SLLN. Theorem 1 follows by scaling Eqs. (A.3) and (A.4) appropriately and applying the Cramer–Wold lemma.

Proof of Theorem 2. In order to prove Theorem 2, we proceed in two stages. We first show that $\hat{\varphi}_{\mathrm{LSDV}}$ is an inconsistent estimator, and derive the inconsistency for $N \to \infty$. We then construct a normalised coefficient statistic corrected for the inconsistency and derive its limiting distribution.

Decompose the vector $y_{i,-1}$ under the null hypothesis that $\varphi = 1$ and $a_i = 0$, as

$$y_{i,-1} = e_T y_{i0} + C_T v_i$$

where C_T is the $(T \times T)$ matrix

$$C_T = \begin{bmatrix} 0 & . & . & 0 \\ 1 & 0 & . & 0 \\ . & . & . & . \\ 1 & 1 & . & 0 \end{bmatrix}$$

and note that, since $Q_T e_T = 0$ (where $Q_T = M_T$), it follows that $Q_T y_{i,-1} = Q_T C_T v_i$. Then Eq. (2) can be written as the ratio

$$\hat{\varphi}_{\mathrm{LSDV}} - 1 = \left[\sum_{i=1}^{N} v_i' C_T' Q_T C_T v_i\right]^{-1} \left[\sum_{i=1}^{N} v_i' C_T' Q_T v_i\right]. \tag{A.5}$$

Dividing both terms of this ratio by σ_v^2 yields a ratio of two summands consisting of quadratic forms in standardised variables which are invariant to y_{i0} and σ_v^2, even in finite samples, thus implying that $\hat{\varphi}_{\mathrm{LSDV}}$ can be used to construct a similar test with respect to the nuisance parameters of Model (1a).

The inconsistency of $\hat{\varphi}_{\mathrm{LSDV}}$ can be derived by writing Eq. (2) based on Eq. (A.5) as

$$\hat{\varphi}_{\mathrm{LSDV}} - 1 = \sum_{i=1}^{N} w_{iT}' \Big/ \sum_{i=1}^{N} D_{iT}' \tag{A.6}$$

where

$$\begin{aligned} w_{iT}' &= \sum_{t=1}^{T} \varepsilon_{it-1} v_{it} - \frac{1}{T} \sum_{t=1}^{T} \varepsilon_{it-1} \sum_{t=1}^{T} v_{it} \\ &= \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} v_{it} v_{is} - \frac{1}{T}\left[\sum_{t=1}^{T} (T-t) v_{it}^2 \right. \\ &\quad \left. + \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} (2T - s - t) v_{it} v_{is}\right] \end{aligned}$$

$$= \sum_{t=1}^{T} \omega_1'(t) v_{iT}^2 + \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} \omega_2'(t,s) v_{it} v_{is}$$

$$D_{iT}' = \sum_{t=1}^{T} \varepsilon_{it-1}^2 - \frac{1}{T}\left(\sum_{t=1}^{T} \varepsilon_{it-1} \right)^2$$

$$= \sum_{t=1}^{T} \delta_1'(t) v_{it}^2 + \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} \delta_2'(t,s) v_{it} v_{is},$$

$$\omega_1'(t) = -(T-t)/T, \quad \omega_2'(t,s) = -(T-s-t)/T,$$

$$\delta_1'(t) = t(T-t)/T \quad \text{and} \quad \delta_2'(t,s) = 2t(T-s)/T.$$

The probability limit of the numerator of Eq. (A.6) is

$$\operatorname*{plim}_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} w_{iT}' = \sum_{t=1}^{T} \omega_1'(t) \left(\operatorname*{plim}_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} v_{it}^2 \right) + \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} \omega_2'(t,s) \left(\operatorname*{plim}_{N\to\infty} \frac{1}{N} \sum_{t=1}^{N} v_{it} v_{is} \right) = -\frac{(T-1)\sigma_v^2}{2}.$$

The probability of the denominator of Eq. (A.6) is

$$\operatorname*{plim}_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} D_{iT}' = \sum_{t=1}^{T} \delta_1'(t) \left(\operatorname*{plim}_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} v_{it}^2 \right) - \frac{1}{T} \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} \delta_2'(t,s) \left(\operatorname*{plim}_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} v_{it} v_{is} \right) = \frac{(T^2-1)\sigma_v^2}{6}$$

and so by the Slutzky theorem, the inconsistency of $\hat{\varphi}_{\mathrm{LSDV}}$ is

$$B_2 = \operatorname*{plim}_{N\to\infty} (\hat{\varphi}_{\mathrm{LSDV}} - 1) = \frac{\operatorname{plim}_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} w_{iT}'}{\operatorname{plim}_{N\to\infty} \frac{1}{N} \sum_{I=1}^{n} D_{iT}'} = \frac{-3}{T+1}. \tag{A.7}$$

Subtracting the inconsistency given by Eq. (A.7) from both sides of Eq. (A.6) yields

$$\hat{\varphi}_{\mathrm{LSDV}} - 1 - B_2 = \frac{\sum_{i=1}^{N} w_{iT}'}{\sum_{i=1}^{N} D_{iT}'} - B_2 = \frac{\sum_{i=1}^{N} x_{iT}'}{\sum_{i=1}^{N} D_{iT}'} \tag{A.8}$$

where $x'_{iT} = w'_{iT} - B_2 D'_{iT}$ is an i.i.d. process, with zero mean, and variance given by

$$\mathrm{Var}(x'_{iT}) = \frac{(T-1)(T-2)(2T-1)}{15T(T+1)}k + \frac{(T-1)(17T^3-44T^2+77T-24)}{60T(T+1)}\sigma_v^4$$

where $\mathrm{Var}(x'_{iT})$ is derived by writing

$$\mathrm{Var}(x'_{iT}) = \mathrm{E}(w'^2_{iT}) + B_2^2\mathrm{E}(D'^2_{iT}) - 2B_2\mathrm{E}(w'_{iT}D'_{iT})$$

and using the following results:

$$\mathrm{E}(w'^2_{iT}) = \frac{(2T-1)(T-1)}{6T}k + \frac{(T-1)(2T^2-4T+3)}{6T}\sigma_v^4,$$

$$\mathrm{E}(D'^2_{iT}) = \frac{(T^2-1)(T^2+1)}{30T}k + \frac{(T^2-1)(T-2)(T^2+1)}{20T}\sigma_v^4,$$

$$\mathrm{E}(w'_{iT}D'_{iT}) = -\frac{(T^2-1)}{12}k - \frac{(T^2-1)(T-2)}{12}\sigma_v^4,$$

where $k = \mathrm{E}(v_{it}^4)$ exists by Assumption (a).

The result of Theorem 2 follows by scaling Eq. (A.8) appropriately, performing manipulations similar to those in Theorem 1, and noting that for the normal distribution, $k/\sigma^4 = 3$.[11]

Proof of Theorem 3. The proof of Theorem 3 follows the same steps as that of Theorem 2. Decompose the vector $y_{i,-1}$, under the null hypothesis that $\beta_i = 0$ and $\varphi = 1$, as

$$y_{i,-1} = e_T y_{i0} + C_T e_T \alpha_i + C_T v_i$$

[11] Relaxing the normality assumption requires the existence of the fourth moment $\mathrm{E}(v_{it}^4)$ in order to derive a limiting distribution of the test statistic given by Theorem 2. Implementation of the test then requires consistent estimation of the kurtosis coefficient under the null hypothesis.

and note that $Q_T Z_T = Q_T(e_T, C_T e_T) = 0$. Then Eq. (2) can be written as

$$\hat{\varphi}_{\mathrm{LSDVT}} - 1 = \left[\sum_{i=1}^{N} v_i' C_T' Q_T C_T v_i\right]^{-1} \cdot \left[\sum_{i=1}^{N} v_i' C_T' Q_T v_i\right]. \tag{A.9}$$

Dividing both the numerator and the denominator of Eq. (A.9) by σ_v^2, yields a ratio of two summands consisting of quadratic forms in standardised variables which are independent of y_{i0}, σ_v^2 and α_i and so $\hat{\varphi}_{\mathrm{LSDVT}}$ can be used to construct a similar test with respect to the nuisance parameters of Model (1b).

The inconsistency of $\hat{\varphi}_{\mathrm{LSDVT}}$ can be derived by writing Eq. (2), according to Eq. (A.9), as

$$\hat{\varphi}_{\mathrm{LSDTV}} - 1 = \sum_{i=1}^{N} w_{iT}'' \Big/ \sum_{i=1}^{N} D_{iT}'' \tag{A.10}$$

where

$$w_{iT}'' = (\tau_T' M_T \tau_T)(\varepsilon_{i,-1}' M_T v_i) - (\varepsilon_{i,-1}' M_T \tau_T)(\tau_T' M_T \varepsilon_{i,-1})$$

$$= \sum_{t=1}^{T} \omega_1''(t) v_{it}^2 + \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} \omega_2''(t,s) v_{it} v_{is},$$

$$\varepsilon_i' = (\varepsilon_{i0}, \ldots, \varepsilon_{iT-1}) \text{ with } \varepsilon_{i0} = 0,$$

$$D_{iT}'' = (\tau_T' M_T \tau_T)(\varepsilon_{i,-1}' M_T \varepsilon_{i,-1}) - (\varepsilon_{i,-1}' M_T \tau_T)(\tau_T' M_T \varepsilon_{i,-1}) \cdot$$

$$= \sum_{t=1}^{T} \delta_1''(t) v_{it}^2 + \sum_{t=1}^{T-1} \sum_{s=t+1}^{T} \delta_2''(t,s) v_{it} v_{is},$$

$$\omega_1''(t) = -\frac{(T-t)(T^2 - 3Tt + 6t^2 - 3t - 1)}{12},$$

$$\omega_2''(t,s) = -\tfrac{1}{12}(T^2-1)(T-t-s)$$

$$+ \tfrac{1}{4}(T+1)[t(T-t) + s(T-s)] - \tfrac{1}{2}ts(2T - t - s),$$

$$\delta_1''(t) = \frac{t(T-t)(T^2 + 3t^2 - 3tT - 1)}{12}$$

and

$$\delta_2''(t,s) = -\tfrac{1}{6}t(T-s)(T^2-3Ts+3ts-1).$$

The probability limit of the numerator of Eq. (A.10) is

$$\begin{aligned}\operatorname*{plim}_{N\to\infty}\frac{1}{N}\sum_{i=1}^{N} w_{iT}'' &= \sum_{t=1}^{T}\omega_1''(t)\left(\operatorname*{plim}_{N\to\infty}\frac{1}{N}\sum_{i=1}^{N} v_{it}^2\right)\\ &\quad+\sum_{t=1}^{T-1}\sum_{s=t+1}^{T}\omega_2''(t,s)\left(\operatorname*{plim}_{N\to\infty}\frac{1}{N}\sum_{t=1}^{N} v_{it}v_{is}\right)\\ &= -\frac{T(T^2-1)(T-2)\sigma_v^2}{24}.\end{aligned}$$

The probability limit of the denominator of Eq. (A.10) is

$$\begin{aligned}\operatorname*{plim}_{N\to\infty}\frac{1}{N}\sum_{i=1}^{N} D_{iT}'' &= \sum_{t=1}^{T}\delta_1''(t)\left(\operatorname*{plim}_{N\to\infty}\frac{1}{N}\sum_{i=1}^{N} v_{it}^2\right)\\ &\quad-\frac{1}{T}\sum_{t=1}^{T-1}\sum_{s=t+1}^{T}\delta_2''(t,s)\left(\operatorname*{plim}_{N\to\infty}\sum_{i=1}^{N} v_{it}v_{is}\right)\\ &= \frac{T(T^2-1)(T^2-2)\sigma_v^2}{180}\end{aligned}$$

and so by the Slutzky theorem, the inconsistency of $\hat{\varphi}_{\mathrm{LSDVT}}$ is

$$B_3 = \operatorname*{plim}_{N\to\infty}(\hat{\varphi}_{\mathrm{LSDVT}}-1) = \frac{\operatorname{plim}_{N\to\infty}(1/N)\sum_{i=1}^{N} w_{iT}''}{\operatorname{plim}_{N\to\infty}(1/N)\sum_{i=1}^{N} D_{iT}''} = -\frac{15}{2(T+2)}. \quad \text{(A.11)}$$

Subtracting Eq. (A.11) from both sides of Eq. (A.10) yields

$$\hat{\varphi}_{\mathrm{LSDVT}} - 1 - B_3 = \frac{\sum_{i=1}^{N} w_{iT}''}{\sum_{i=1}^{N} D_{iT}''} - B_3 = \frac{\sum_{i=1}^{N} x_{iT}''}{\sum_{i=1}^{N} D_{iT}''} \quad \text{(A.12)}$$

where $x_{iT}'' = w_{iT}'' - B_3 D_{iT}''$ is an i.i.d. process, with zero mean, and variance given by

$$\begin{aligned}\operatorname{Var}(x_{iT}'') &= \frac{T(T^2-1)(T-2)(37T^4-302T^3+384T^2+1934T-2701)}{120{,}960(T+2)}k\\ &+\frac{T(T^2-1)(T-2)(193T^5-950T^4-2766T^3-1576T^2-12751T+16206)}{241{,}920(T+2)}\sigma_V^4\end{aligned}$$

where $\mathrm{Var}(x''_{iT})$ is derived by writing

$$\mathrm{Var}(x''_{iT}) = \mathrm{E}(w''^{2}_{iT}) + B_3^2\mathrm{E}(D''^{2}_{iT}) - 2B_3\mathrm{E}(w''_{iT}D''_{iT})$$

and using the following results

$$\mathrm{E}(w''^{2}_{iT}) = \sum_{t=1}^{T}\omega''_1(t)^2k + \sum_{t=1}^{T-1}\sum_{s=t+1}^{T}[2\omega''_1(t)\omega''_1(s) + \omega''_2(t,s)^2]\sigma_v^4$$

$$= \frac{T(T-2)(T^2-1)(29T^3-47T^2-23T+59)}{15{,}120}k$$

$$+ \frac{T(T^2-1)(T-2)(56T^4-167T^3-226T^2+131T-354)}{30{,}240}\sigma_v^4,$$

$$\mathrm{E}(D''^{2}_{iT}) = \sum_{t=1}^{T}\delta''_1(t)^2k + \sum_{t=1}^{T-1}\sum_{s=t+1}^{T}[2\delta''_1(t)\delta''_1(s) + \delta''_2(t,s)^2]\sigma_v^4$$

$$= \frac{T(T^2-1)(T^2-2)(T^2-5)(T^2+5)}{30{,}240}k$$

$$+ \frac{T(T^2-1)(T^2-2)(13T^5-30T^4+10T^3-23T+750)}{302{,}400}\sigma_v^4,$$

and

$$\mathrm{E}(w''_{iT}D''_{iT}) = \sum_{t=1}^{T}\omega''_1(t)\delta''_1(t)k + \sum_{t=1}^{T-1}\sum_{s=t+1}^{T}[\omega''_1(t)\delta''_1(s)$$

$$+ \omega''_1(s)\delta''_1(t) + \omega''_2(t,s)\delta''_2(t,s)]\sigma_v^4$$

$$= -\frac{T(T^2-1)(T^2-2)(14T^3-15T^2-14T-33)}{60{,}480}k$$

$$-\frac{T(T^2-1)(T^2-2)(14T^4-42T^3+31T^2+42T+99)}{60{,}480}\sigma_v^4.$$

The result of Theorem 3 follows by the same arguments as in Theorem 2. □

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